

Assignment 2

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Preparation

```
# packages
library("brms")
library("ggplot2")
library("tidyverse")

# data
load("data/income.RData")
```

```
data: income.RData
```

```
head(income)
```

```
##      ls emotst inc_hh emotst_cat
## 1 3.8    6.0     7      high
## 2 1.0    3.0     6      low
## 3 5.0    4.0    10      low
## 4 4.2    3.5     7      low
## 5 4.4    6.0     2      high
## 6 3.8    6.5     7      high
```

The data set contains four variables

Variable	Description
ls	Life satisfaction (1-5)
emotst	Emotional stability (1-7)
emotst_cat	Emot. stability (high-low)
inc_hh	Household income (in 10.000)

Exercise 1

Fit the following Bayesian model to assess the effect of household income on life satisfaction.

```
# Casewise exclusion of missing data
income <- na.exclude(income)

# This function can be used to figure out the structure of the
# priors for the sepcific model including names of the predictors
# default_prior(ls ~ inc_hh,
#               data = income)

bprior <- c(brms::prior(normal(3, 2), class = Intercept)
            , brms::prior(normal(0, 1), class = b, coef = inc_hh)
            , brms::prior(normal(0, 2.5), class = sigma))
```

```

model.1 <- brm(ls ~ 1 + inc_hh
              , data = income
              , prior = bprior
              , silent = 2
              , refresh = 0)

## Trying to compile a simple C file

## Running /usr/lib/R/bin/R CMD SHLIB foo.c
## using C compiler: 'gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0'
## gcc -std=gnu23 -I"/usr/share/R/include" -DNDEBUG -I"/home/julia/R/x86_64-pc-linux-gnu-library/4.5/
## In file included from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/StanHeaders/include/stan/math/pr
##                  from <command-line>:
## /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10:
##   679 | #include <cmath>
##       |          ^~~~~~
## compilation terminated.
## make: *** [/usr/lib/R/etc/Makeconf:202: foo.o] Error 1

```

- a) Consider the priors. Do you think they are a good choice? Run a prior prediction and change the priors if necessary.
- b) Fit the model with the adjusted priors. Ensure convergence and visualize the posterior distributions of all parameters. Interpret the results.

Solution

- Y_i : Life satisfaction of the i th person
- x_i : Income

$$Y_i \sim \text{Normal}(\mu_i, \sigma^2)$$

$$\mu_i = \beta_0 + x_i \beta_1$$

1a. The `ls`-values are between 1 and 5, so effects cannot be larger than one step on the life satisfaction scale. The mean is probably around 3. Sigma, the noise standard deviation can also not be bigger than 2-3 because of the range restriction of the data.

```

dat <- data.frame(ls = rnorm(nrow(income), 2.5, 1), inc_hh = sample(1:12, size = nrow(income), replace =
model.prior.pred <- brm(ls ~ 1 + inc_hh
                       , data = dat
                       , prior = bprior
                       , sample_prior = "only"
                       , silent = 2
                       , refresh = 0
                       , cores = 2)

```

```

## Trying to compile a simple C file

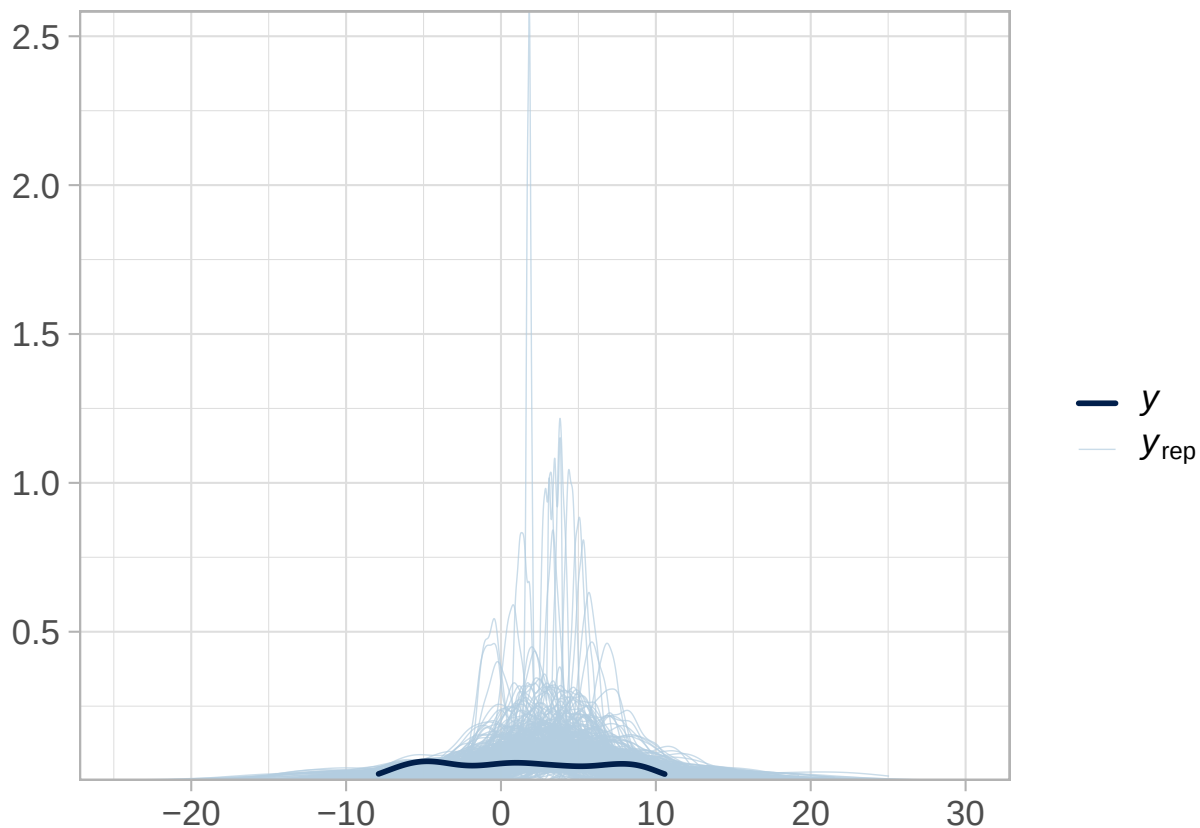
## Running /usr/lib/R/bin/R CMD SHLIB foo.c
## using C compiler: 'gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0'
## gcc -std=gnu23 -I"/usr/share/R/include" -DNDEBUG -I"/home/julia/R/x86_64-pc-linux-gnu-library/4.5/
## In file included from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/StanHeaders/include/stan/math/pr

```

```
##           from <command-line>:
## /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10:
## 679 | #include <cmath>
##     |         ^~~~~~
## compilation terminated.
## make: *** [/usr/lib/R/etc/Makeconf:202: foo.o] Error 1
```

```
prior.predictions <- posterior_predict(model.prior.pred)

bayesplot::ppc_dens_overlay(prior.predictions[1,], prior.predictions[1:500,], trim = T) +
  # xlim(-5, 10) +
  theme_light(base_size = 16)
```



Life satisfaction is on a scale from 1 to 5, so the range should be smaller.

1b.

```
cprior <- c(brms::prior(normal(3, 1), class = Intercept)
  , brms::prior(normal(0, .5), class = b, coef = inc_hh)
  , brms::prior(normal(0, 1), class = sigma))
model.2 <- brm(ls ~ 1 + inc_hh
  , data = income
  , prior = cprior
  , silent = 2
  , refresh = 0
  , cores = 4)
```

```
## Trying to compile a simple C file
```

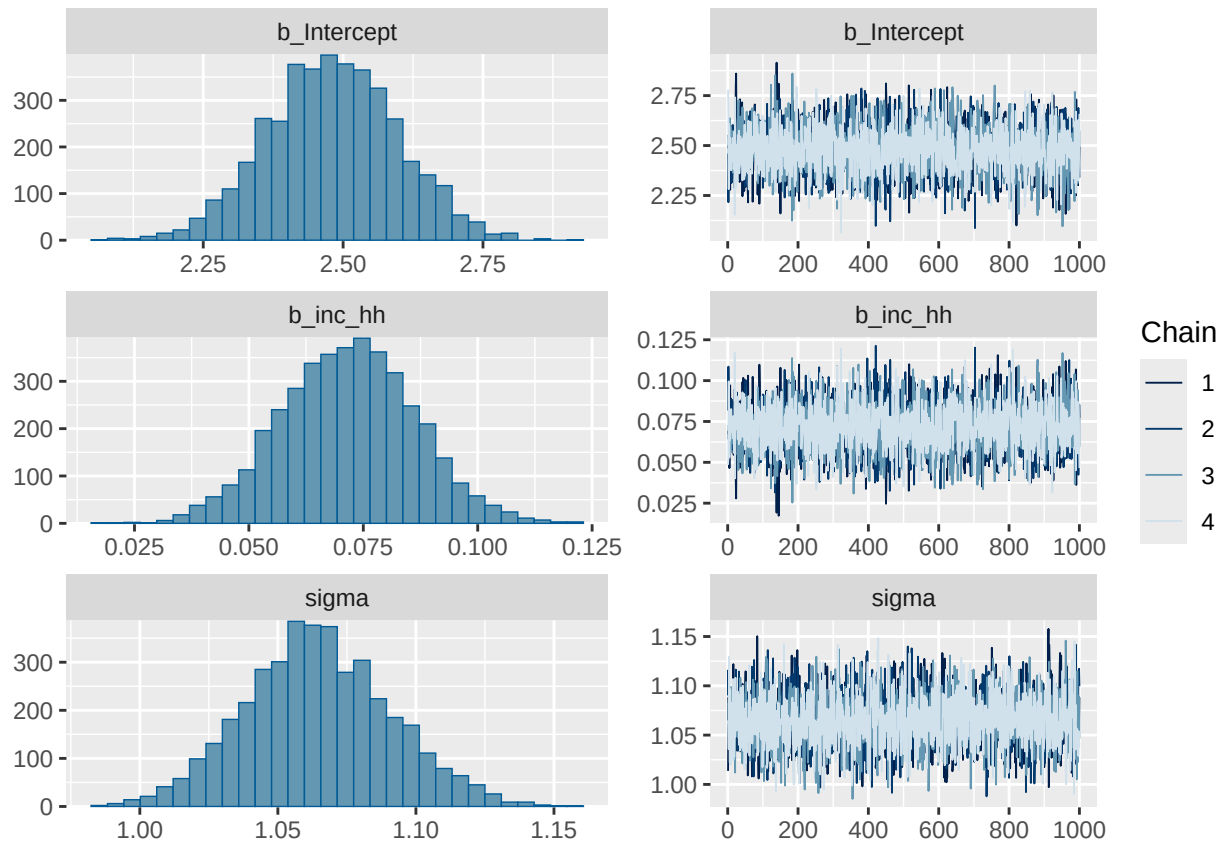
```
## Running /usr/lib/R/bin/R CMD SHLIB foo.c
```

```
## using C compiler: 'gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0'
## gcc -std=gnu23 -I"/usr/share/R/include" -DNDEBUG -I"/home/julia/R/x86_64-pc-linux-gnu-library/4.5/
## In file included from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Core:19,
## from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Dense:1,
## from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/StanHeaders/include/stan/math/pr
## from <command-line>:
## /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10:
## 679 | #include <cmath>
## | ^~~~~~
## compilation terminated.
## make: *** [/usr/lib/R/etc/Makeconf:202: foo.o] Error 1
```

model.2

```
## Family: gaussian
## Links: mu = identity
## Formula: ls ~ 1 + inc_hh
## Data: income (Number of observations: 828)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Regression Coefficients:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      2.48      0.12    2.25    2.71 1.00    3984    3177
## inc_hh          0.07      0.01    0.04    0.10 1.00    3901    3042
##
## Further Distributional Parameters:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma      1.06      0.03    1.01    1.12 1.00    3626    2676
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

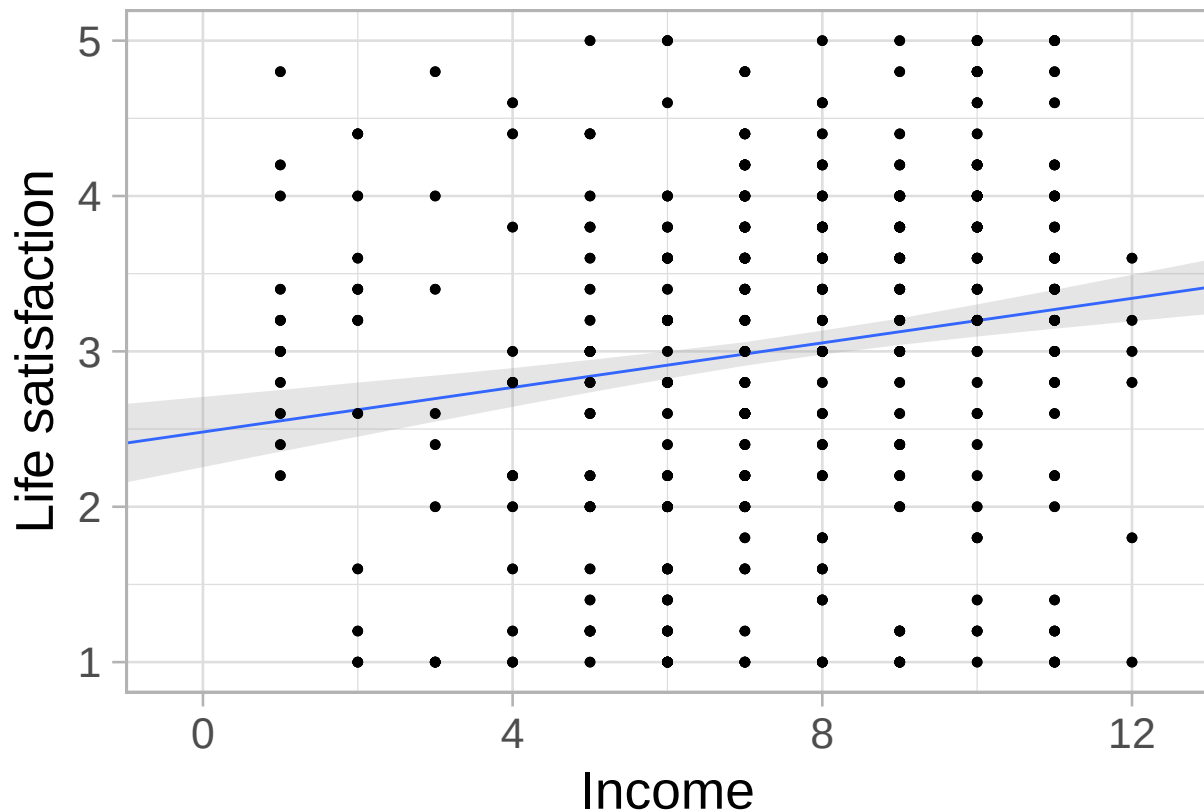
plot(model.2)



Convergence is good. $\hat{R}$ and ESS are in the range and the caterpillars are hairy.

None of the posterior credible intervals includes 0. For an increase of 10.000 Euros of household income there is an increase of life satisfaction of about 0.07 points.

```
## Neue Daten werden erstellt, um die Regressionsgerade zu zeichnen
nd <- tibble(inc_hh = -1:13, 2)
## Regression wird auf die neuen Daten angewendet
f <- fitted(model.2, newdata = nd) %>%
  as_tibble() %>%
  bind_cols(nd)
income %>%
  ggplot(aes(x = inc_hh)) +
  geom_smooth(data = f,
    aes(y = Estimate, ymin = Q2.5, ymax = Q97.5),
    stat = "identity",
    alpha = 1/4, linewidth = 1/2) +
  geom_point(aes(y = ls),
    size = 2/3) +
  scale_x_continuous("Income", expand = c(0, 0)) +
  ylab("Life satisfaction") +
  theme_light(base_size = 20, )
```



c) Conduct a sensitivity analysis by fitting the model using three different priors.

Solution 1c.

```
cprior_a <- c(brms::prior(normal(3, .5), class = Intercept)
             , brms::prior(normal(0, .5), class = b, coef = inc_hh)
             , brms::prior(normal(0, .5), class = sigma))
model.2_a <- brm(ls ~ 1 + inc_hh
                , data = income
                , prior = cprior_a
                , silent = 2
                , refresh = 0
                , cores = 4)
```

```
## Trying to compile a simple C file
```

```
## Running /usr/lib/R/bin/R CMD SHLIB foo.c
```

```
## using C compiler: 'gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0'
```

```
## gcc -std=gnu23 -I"/usr/share/R/include" -DNDEBUG -I"/home/julia/R/x86_64-pc-linux-gnu-library/4.5/
```

```
## In file included from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Core:19,
```

```
## from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Dense:1,
```

```
## from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/StanHeaders/include/stan/math/pr
```

```
## from <command-line>:
```

```
## /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10:
```

```
## 679 | #include <cmath>
```

```
## | ^~~~~~
```

```
## compilation terminated.
```

```
## make: *** [/usr/lib/R/etc/Makeconf:202: foo.o] Error 1
```

```
model.2_a
```

```
## Family: gaussian
## Links: mu = identity
## Formula: ls ~ 1 + inc_hh
## Data: income (Number of observations: 828)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Regression Coefficients:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      2.48      0.11    2.27    2.70 1.00     4139     3082
## inc_hh          0.07      0.01    0.04    0.10 1.00     4104     3103
##
## Further Distributional Parameters:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma      1.06      0.03    1.01    1.12 1.00     3873     2659
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
cprior_b <- c(brms::prior(normal(3, 2), class = Intercept)
              , brms::prior(normal(0, .5), class = b, coef = inc_hh)
              , brms::prior(normal(0, 2), class = sigma))
model.2_b <- brm(ls ~ 1 + inc_hh
                 , data = income
                 , prior = cprior_b
                 , silent = 2
                 , refresh = 0
                 , cores = 4)
```

```
## Trying to compile a simple C file
```

```
## Running /usr/lib/R/bin/R CMD SHLIB foo.c
## using C compiler: 'gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0'
## gcc -std=gnu23 -I"/usr/share/R/include" -DNDEBUG -I"/home/julia/R/x86_64-pc-linux-gnu-library/4.5/
## from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Core:19,
## from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Dense:1,
## from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/StanHeaders/include/stan/math/pr
## from <command-line>:
## /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10:
## 679 | #include <cmath>
## |      ^~~~~~
## compilation terminated.
## make: *** [/usr/lib/R/etc/Makeconf:202: foo.o] Error 1
```

```
model.2_b
```

```
## Family: gaussian
## Links: mu = identity
## Formula: ls ~ 1 + inc_hh
## Data: income (Number of observations: 828)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
```

```
## Regression Coefficients:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      2.48      0.12    2.25    2.71 1.00     3924     2805
## inc_hh         0.07      0.01    0.04    0.10 1.00     3869     2878
##
## Further Distributional Parameters:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma         1.07      0.03    1.02    1.12 1.00     3951     3128
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
cprior_c <- c(brms::prior(normal(3, 1), class = Intercept)
              , brms::prior(normal(-1, 1), class = b, coef = inc_hh)
              , brms::prior(normal(0, 1), class = sigma))
model.2_c <- brm(ls ~ 1 + inc_hh
                 , data = income
                 , prior = cprior_c
                 , silent = 2
                 , refresh = 0
                 , cores = 4)
```

```
## Trying to compile a simple C file
```

```
## Running /usr/lib/R/bin/R CMD SHLIB foo.c
## using C compiler: 'gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0'
## gcc -std=gnu23 -I"/usr/share/R/include" -DNDEBUG -I"/home/julia/R/x86_64-pc-linux-gnu-library/4.5/
## In file included from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/StanHeaders/include/stan/math/pr
##                  from <command-line>:
## /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10:
## 679 | #include <cmath>
##      |          ^~~~~~
## compilation terminated.
## make: *** [/usr/lib/R/etc/Makeconf:202: foo.o] Error 1
```

```
model.2_c
```

```
## Family: gaussian
## Links: mu = identity
## Formula: ls ~ 1 + inc_hh
## Data: income (Number of observations: 828)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Regression Coefficients:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      2.49      0.11    2.27    2.70 1.00     3851     2873
## inc_hh         0.07      0.01    0.04    0.10 1.00     3962     2864
##
## Further Distributional Parameters:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma         1.07      0.03    1.01    1.12 1.00     4203     2971
##
```



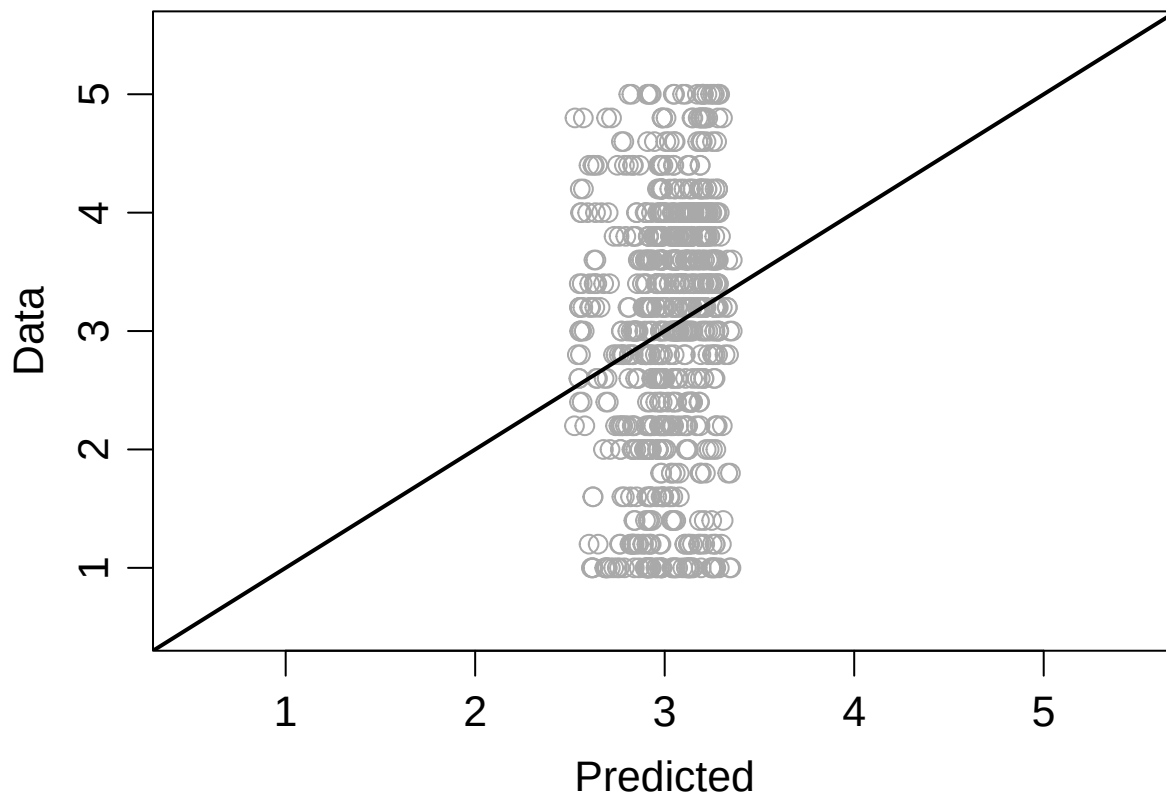
```
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

d) Run a posterior prediction. Visualize the results.

Solution 1d.

```
post.pred.1 <- posterior_predict(model.2)

par(mar = c(4,4,.5,.5), cex = 1.3, mgp = c(2, .7, 0))
post.pred.mean.1 <- colMeans(post.pred.1)
plot(post.pred.mean.1
     , income$ls
     , ylab = "Data", xlab = "Predicted"
     , pch = 1, col = "darkgray"
     , ylim = c(.5, 5.5), xlim = c(.5, 5.5))
abline(0, 1, lwd = 2)
```



```
cor(post.pred.mean.1, income$ls)
```

```
## [1] 0.171397
```